

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Theory Examination Nov/Dec 2016

PH305 Pharmaceutical Microbiology

Date: 23/11/16, Wednesday Time: 10:00 a.m to 1:00 p.m Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION-I

Q 1 Attempt any TWO of the following

- A (1) What do you understand by the following terms: 04
Disinfection, Sterilisation, Antiseptic, Bacteriostatic
- (2) What properties are required for an ideal Disinfectant? 03
- (3) Classify the various sterilisation processes. 03
- B (1) What mechanism is involved in sterilisation by: 04
Moist heat, Dry Heat, Radiation, Gasous/Ethylene Oxide
- (2) How are disinfectants are evaluated? 03
- (3) Write the effects of pH, temperature and concentration of disinfectant on the disinfectant activity. 03
- C (1) Define following terms: 04
Z value, Thermal Death Time, Decimal Reduction time or D value, F value
- (2) Enlist the various factors which influence the efficiency of Disinfectants. 03
- (3) Discuss the factors affecting sterilisation process. 03

Q 2 Attempt any TWO of the following

- A (1) Explain principles of sterility testing of pharmaceuticals. Give the importance of performing sterility test. 04
- (2) Give composition of media used for sterility testing and use of each ingredient. 03
- (3) Write a short note on sterility of eye preparation. 03
- B (1) Define and discuss the term microbiological assay. 04
- (2) Write a note on microbiological assay of calcium pantothenate. 03
- (3) Describe turbidimetric microbiological assay. 03

- C** (1) How will you perform test for sterility of the following preparations? **04**
- i) surgical gauge
 - ii) camphor injection
 - iii) talc powder
 - iv) dextrose saline injection in 400 ml bottle
- (2) Discuss the principle underlying the microbiological assay of antibiotics. **03**
- (3) What is Pyrogen. Write a short note on Pyrogen with respect to Chemical composition. **03**

SECTION- II

Q 3 Attempt any TWO of the following

- A** 1. Write a note on spirochetes. **5**
2. Describe in detail factor affecting growth of microorganism . **5**
- B** Explain in detail microbial assay of antibiotics. **10**
- C** Write a note on Staining methods of microbes. **10**

Q 4 Attempt any TWO of the following

- A** 1. What is the scope of microbiology in production of medicine? **5**
2. Differentiate : Light Microscope & Electron Microscope **5**
- B** 1. Explain in detail – Cultivation methods of microorganisms **5**
2. Write a note on bacterial cell wall **5**
- C** 1. Differentiate: Selective Media & Diagnostic Media **5**
2. Write a note on – Flagella and Pili **5**
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